Project Documentation: India Electricity Consumption Analysis (COVID-19 Impact)

1. Project Overview

This project provides an end-to-end data engineering and visualization solution to analyze the shift in India's power consumption before, during, and after the 2020 COVID-19 lockdowns. The analysis tracks the transition from a pre-pandemic baseline through the industrial shutdown and maps the subsequent economic recovery across different states and geographic regions.

Tech Stack: * **Data Processing:** Python (Pandas)

- **Database:** MySQL 8.0
- **Visualization:** Tableau Public
- **Frontend UI:** Flask (Python), HTML, Tailwind CSS

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2. Step-by-Step Development Procedure

Phase 1: Data Collection & Extraction

- **Dataset Acquisition:** Sourced a comprehensive time-series dataset detailing daily state-wise power consumption (in Mega Units) from 2013 to 2024.
- **Timeline Definition:** Isolated the project scope to focus strictly on the period between **January 1, 2019, and December 5, 2020**, to capture the complete pre-lockdown, strict lockdown, and recovery phases.

Phase 2: Data Preparation (ETL Process)

- **Data Transformation:** Developed a Python script utilizing the Pandas library to clean the raw dataset.
- **Dimensional Remodeling:** Converted the dataset from a "wide" format (states as individual columns) to a "long" format (Dates, State, Consumption) using the pd.melt() function to make it compatible with relational databases and BI mapping tools.
- **Data Standardization:** Standardized anomalous geographic data strings (e.g., mapping "Pondy" to "Puducherry" and combining "DNH" and "DD") to align with standard geographic coordinate systems.

Phase 3: Database Storage & SQL Operations

- **Schema Creation:** Engineered a normalized relational database schema (electricity_final) in MySQL to handle high-volume time-series data.
- **Data Ingestion:** Executed high-speed bulk data imports, successfully loading over 23,000+ records.
- **SQL Analytics:** Performed validation queries utilizing aggregate functions (SUM, GROUP BY, BETWEEN) to verify regional consumption drops during the April 2020 lockdown period.

Phase 4: Tableau Data Visualization

- **Data Connection:** Linked the cleaned dataset to Tableau and assigned correct Geographic Roles (State/Province) to generate spatial maps of India.
- **Calculated Fields:** Programmed custom logical fields to segment the data:
    1. Lockdown Phase: An IF-ELSE statement categorizing dates into Pre-Lockdown, Lockdown, and Recovery.
    2. Region Group: A geographic mapping field grouping 30+ individual states into 5 main regions (Northern, Southern, Eastern, Western, North-Eastern).
- **Unique Visualizations:** Designed three core analytical charts:
    1. **Scenario 1 (National Trend):** A time-series Area Chart visualizing the massive dip in national power usage from March to June 2020.
    2. **Scenario 2 (Regional Variations):** A stacked Bar Chart highlighting which industrial sectors (e.g., Western vs. Northern India) experienced the sharpest declines.
    3. **Scenario 3 (Recovery):** A choropleth Map of India comparing pre-pandemic baselines to late-2020 usage to prove economic recovery.

Phase 5: Dashboarding & Storytelling

- **Responsive Design:** Assembled the three visualizations into a unified, responsive dashboard precisely sized at **1200x800 pixels** for web embedding.
- **Interactive Filtering:** Implemented cross-dashboard filter actions, allowing users to click on a specific state on the map to dynamically update the corresponding trend lines and regional metrics.
- **Data Storytelling:** Constructed a 3-scene Tableau Story to guide the viewer chronologically through the data narrative: The Crash, The Regional Divide, and The Recovery.

Phase 6: Web Integration (Flask)

- **Application Architecture:** Built a lightweight Python web server using the Flask framework to host the final analysis.
- **UI/UX Design:** Engineered a modern, "SaaS-style" frontend utilizing Tailwind CSS. The design features a dark-mode cyberpunk aesthetic, glassmorphism UI cards, and dynamic KPI trackers.
- **Embed Implementation:** Seamlessly integrated the 1200x800 Tableau Story into the center of the web application, ensuring a zero-gap, interactive user experience directly from the browser.

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3. Performance Testing Metrics

To ensure the solution was robust and scalable, the following metrics were recorded during the testing phase:

- **Amount of Data Rendered to DB:** 23,000+ individual daily consumption records.
- **Number of Unique Visualizations/Graphs:** 3 highly customized charts (Area, Bar, Map).
- **Number of Calculation Fields:** 3 custom fields (Lockdown Phase, Region Group, % Change in Consumption).
- **Utilization of Data Filters:** Dynamic, dashboard-wide date sliders and action-based geographic filters were successfully implemented without latency.